IDE BOOTCAMP 2026

JIVA: REAL-TIME HEALTHCARE COORDINATION

PROJECT IDENTIFICATION & CORE THESIS

- **Problem Statement Title:** Real-Time Emergency Healthcare Coordination & Dynamic Hospital Routing
- **Theme:** Healthcare & MedTech / Cloud-Native Emergency Mesh
- **PS Category:** Software (Event-Driven Healthcare Mesh)
- **Team Name:** Monochrome
- **Team Members:** Yukith M Joseph • Pranathi Nagabhushana
- **Solution Summary:** An event-driven layer maintaining a shared, timestamped care state for patients, ambulances, and hospitals, routing via explicit hospital acceptance and dynamically rerouting when state changes.
- **Verification Status:** Fully functional local prototype verified on synthetic Bengaluru data (124/124 checks pass). AWS architecture modeled in CDK (not deployed).

JIVA: CLOSED-LOOP HEALTHCARE COORDINATION

PROPOSED SOLUTION: The Closed-Loop Hospital Acceptance & Dynamic Rerouting Engine

1. Emergency Intake
Patient / Paramedic logs incident
Clinical requirements tagged
Canonical event ingested

→

2. Feasibility Gate
Candidate hospitals evaluated
Determinism & freshness check
Identifies eligible facilities

→

3. Acceptance SLA
Targeted request dispatched
Case-scoped confirmation query
Strict timeout countdown

→

4. Explicit Response
Mandatory active confirmation:
[ACCEPTED] [LIMITED] [REJECTED] [UNAVAILABLE]
Silence expires to UNKNOWN

→

5. Dynamic Reroute
Confirmed unit dispatched
Real-time GPS telemetry sync
Reroutes if hospital degrades

✗ EXISTING REALITY: Static & Unverified Assumptions

- **Assumed Readiness:** Directory listings prove a hospital exists, not that an ICU bed or specialist is free right now.
- **Fragmented Phone Calls:** Paramedics negotiate en-route over phone; no shared timestamped picture.
- **Fatal ER Diversion:** Ambulance arrives unannounced at saturated emergency room → turned away at the door.
- **Static Destination Lock:** Destination is chosen once at dispatch; zero awareness if conditions degrade en-route.

✓ JIVA SOLUTION: Closed-Loop Event-Driven Coordination

- **Explicit Confirmation:** Hospitals must actively confirm acceptance; unconfirmed requests deterministically expire to UNKNOWN.
- **Live Dynamic Rerouting:** If an accepted hospital becomes UNAVAILABLE en-route, JIVA immediately triggers dynamic rerouting.
- **Zero EHR Intrusion:** Lightweight web overlay across 4 role-scoped apps (:5174-:5177); zero costly HIS integrations.
- **Deterministic Feasibility:** 100% deterministic rule verification & SHA-256 event audits; generative AI strictly advisory.

TECHNICAL APPROACH & ARCHITECTURE

SYSTEM ARCHITECTURE: Hybrid Local-to-Cloud Event-Driven Mesh

CLIENT TIER
React 19 / Vite

[Patient Portal :5174]

[Paramedic App :5175]

[Hospital ED :5176]

[Coordinator :5177]

INGESTION & AUTH
API Gateway / REST

- Zod Schema Validation
- Strict Payload Filtering
- Cognito / Role Tokens
- Policy Hash (hashPolicy)
- Port-Keyed Isolation

EVENT MESH & DOMAIN LOGIC
Amazon EventBridge + Lambda / Node.js

- **Canonical Event Bus:** Broadcasts validated lifecycle events
- **State Transition Engine:** Tracks PENDING → ACCEPTED / REJECTED
- **Dynamic Reroute Logic:** Auto-triggers recalculation on status change
- **Deterministic Shadow Engine:** Parallel feasibility, freshness & provenance check
- **Amazon Bedrock (Advisory):** Strictly clinical summaries; ZERO routing authority

PERSISTENCE & SPATIAL
DynamoDB & MapLibre GL

- **Operational State Store:** In-memory locally (<50ms) / DynamoDB cloud
- **SHA-256 Audit Ledger:** Immutable forensic history of every state change
- **Spatial Map Engine:** MapLibre GL client-side WebGL vector rendering
- **Routing Engine Fallback:** Valhalla/OSRM; failures yield UNKNOWN (not 0)

IMPLEMENTATION METHODOLOGY & AUTOMATED VERIFICATION BASELINE

Phase 1: Ingestion
Zod schemas, event bus, & role contracts

Phase 2: Acceptance
Case-scoped requests & timeout SLA

Phase 3: Dynamic Reroute
Auto-rerouting upon hospital degradation

Phase 4: Multi-App Mesh
4 role-scoped apps + Live Demo portal

✓ 124/124 Runtime Checks Passing
Flagship simulation & parity verified

✓ 25 Unit Test Suites Verified
Domain, transit, & auth coverage

✓ 7 Lambda Bundles & CDK Synth
esbuild compiled & cleanly synthesized

FEASIBILITY AND VIABILITY

Operational Feasibility
Zero EHR Intrusion

- **Lightweight Web Overlay:** Runs in any browser without costly integration into proprietary hospital EHR/HIS systems.
- **Zero Hardware Lock-In:** Compatible with any ambulance tablet, phone, or ER desktop; ECG IoT simulated via JSON.
- **Sub-50ms Processing:** Synchronous in-memory pipeline ensures instant operational response times.

Technical Feasibility
Offline-First to Cloud

- **Self-Contained Local Prototype:** 100% offline-capable on port-isolated Vite servers for field / hackathon resilience.
- **CDK Serverless Architecture:** Clean 1:1 mapping from local components to AWS EventBridge, Lambda, and DynamoDB.
- **Shadow Engine Verification:** Runs parallel deterministic checks with 0 regression disagreements.

Economic Viability
High-Impact Cost Efficiency

- **Serverless Pay-Per-Event:** Negligible idle cost; scales dynamically with emergency call volume at fractions of a cent.
- **Prevents Fatal Diversions:** Eliminates catastrophic costs associated with secondary transfers and ER turnaways.
- **Resource Optimization:** Reduces wasted ambulance mileage and enables prepared hospital intake.

RISK VS. STRATEGY MITIGATION MATRIX

⚠ **[RISK] Public Directory Blind Spots:** Govt hospital registries list facilities but lack live ICU / bed status.	✓ **[STRATEGY] Honest Availability Semantics:** Unconfirmed capacities strictly flagged UNKNOWN; explicit acceptance is mandatory.
⚠ **[RISK] Urban Transit Cellular Drops:** Mobile connectivity fluctuates during high-speed ambulance transit.	✓ **[STRATEGY] Offline Caching & Reconnection:** Local client state caching, auto Socket.IO backoff; keeps last GPS with UNKNOWN ETA.
⚠ **[RISK] In-Memory State Volatility:** Prototype runs in memory; process crash resets active emergency session.	✓ **[STRATEGY] Enterprise Persistence Path:** Migration roadmap mapped to transactional SQLite locally and DynamoDB single-table cloud.
⚠ **[RISK] Generative AI Hallucinations:** Unbounded LLMs could hallucinate clinical triaging or routing destinations.	✓ **[STRATEGY] Strict Advisory AI Sandboxing:** Amazon Bedrock strictly sandboxed to read-only summaries; routing is 100% deterministic code.

IMPACT AND BENEFITS

Patients & Families
Transparent, Honest Emergency Transit

- Honest pre-arrival travel ETAs (never fake 0-min ETAs).
- Eliminates traumatizing ER diversions & secondary hospital turnaways.
- Live transparent care status tracking during critical care intervals.

Paramedic Crews
Confirmed Destinations & Zero Phone Chaos

- Guaranteed confirmed hospital bed before ambulance leaves the scene.
- Automated hands-free dynamic reroute alerts if receiving ER status degrades.
- Replaces chaotic en-route phone calls with structured digital handover.

Clinical & Social Impact
Compressing Door-to-Needle Intervals

Dramatically reduces critical delays in acute stroke, STEMI, and trauma by eliminating secondary hospital transfers.
(Saves ~1.9M brain neurons per minute saved in ischemic stroke).

Economic & Urban Efficiency
Minimizing Fuel & Congestion Waste

Dynamic shortest-feasible-path emergency routing eliminates circular transit, wasted vehicle idling, and citywide traffic disruption while optimizing ER throughput.

Hospital Emergency Staff
Structured Pre-Arrival Telemetry & Prep

- Structured pre-arrival vitals enable advance trauma & cath lab preparation.
- One-click intake control: ACCEPT, LIMITED, or REJECT based on live capacity.
- Prevents ER bottleneck congestion and uncoordinated ambulance flooding.

Regional Care Coordinators
District-Wide Situational Awareness

- Real-time operational dashboard across regional hospital beds & dispatches.
- Active monitoring of acceptance requests & expiry SLA countdowns.
- Dynamic cross-facility load balancing during mass-casualty incidents.

Forensic Governance
Tamper-Evident SHA-256 Ledger

Cryptographic policy hashing (hashPolicy) and immutable SHA-256 event audit trails provide forensic accountability for clinical governance and medico-legal auditability.

RESEARCH AND REFERENCES

Public Healthcare Registries
Data Grounding & Facility Identity

- **Karnataka State Health & Family Welfare Services:** https://hfw.karnataka.gov.in — official registry for state healthcare facilities.
- **BBMP Open Hospital Directory:** Bruhat Bengaluru Mahanagara Palike urban healthcare facility coordinates and publicly listed specialties.
- **7 Canonical Bengaluru Facilities:** Ground-truth benchmark dataset derived from open government registries; capacity marked UNKNOWN.

Healthcare & Facility Standards
Data Modeling & Nomenclature

- **Registry of Hospitals in Network of Insurance (ROHINI):** Insurance Information Bureau of India (IIB) (https://iib.gov.in) — facility identification standards.
- **HL7 FHIR Release 4:** https://hl7.org/fhir — foundational standards for timestamped clinical resource representation and event provenance.
- **Emergency Classification:** Standardized care requirement taxonomies for stroke, cardiac arrest, trauma, and burn cases.

Spatial Engine & Mapping Framework
Geospatial Routing Architecture

- **OpenStreetMap (OSM) Contributors:** https://www.openstreetmap.org — open-access topological road network vector graph.
- **MapLibre GL:** https://maplibre.org — hardware-accelerated WebGL client-side vector map renderer.
- **Valhalla & OSRM Routing Engines:** High-performance multimodal matrix calculation with safe fallback to UNKNOWN ETA.

Cloud & Architectural Standards
Event-Driven & AI Guardrails

- **AWS Well-Architected Framework:** Healthcare Industry Lens & Event-Driven Architecture (Amazon EventBridge, Lambda, DynamoDB).
- **DynamoDB Single-Table Design:** Low-latency sub-50ms operational state persistence architecture.
- **Advisory AI Guardrails:** Amazon Bedrock sandboxed to read-only clinical summaries; strictly excluded from routing logic.

PROJECT REPOSITORY & VERIFIED ARTIFACTS

- **GitHub Repository:** https://github.com/Zynx095/Jiva — Monorepo containing full API, 4 web apps, demo orchestrator, and CDK.
- **Verification Baseline:** 25 automated test suites pass • 124 runtime checks verified • 7 Lambda bundles compiled • CDK synth passes.